

암생물학(이) (LS414-01)

학기	2025년 2학기
담당교수	정찬
이수구분	교선평수
학점	3.0
강의시간	화14:30-16:00(E7 - L20), 목14:30-16:00(E7 - L20)
강의실	E5 213 (교수실)
소속	기초학부

교과개요

This lecture will cover current concepts and knowledge of cancer, including cancer pathogenesis and treatments. It will introduce how various genetic and molecular changes transform normal cells into malignant cancer cells as well as develop cancer. Topics include the natural history of cancer, oncogenes, tumor suppressors, cancer-causing viruses, and current therapeutic approaches to cancer treatment.

교과목표

Students will gain knowledge of the cellular and molecular basis of cancers, learn techniques commonly used in cancer biology, and sharpen their critical thinking skills. Students will be able to describe the six hallmarks of cancer, explain the types of gene mutations and how these mutations can contribute to cancer formation. Also, they will be able to describe why an oncogene and tumor suppressor gene are important in cancer development and how they help cancer to escape cell death. Finally, students will be able to understand current cancer treatments and develop new approaches in cancer treatment.

주차별 강의계획

- 1주차:The Nature of Cancer
- 2주차:Tumor Viruses
- 3주차:Oncogenes
- 4주차:Cell Proliferation in cancer (Growth Factors, Receptors)
- 5주차:Cell Signaling in cancer
- 6주차:Tumor Suppressor Genes
- 7주차:pRb and p53
- 8주차:Midterm examination.
- 9주차:Eternal life
- 10주차:Carcinogenesis
- 11주차:Genomic Instability in cancer development
- 12주차:Tumor microenvironment
- 13주차:Moving out
- 14주차:Tumor immune system
- 15주차:Tumor treatment (We can-cer vival)
- 16주차:Final examination

학습윤리

-Academic Dishonesty Any of the following acts, when committed by a student, shall constitute academic dishonesty: • Cheating: Intentionally using or attempting to use unauthorized materials, information, or study

aids in any academic exercise. • Fabrication: Intentional and unauthorized falsification or invention of any information or citation in an academic exercise. • Facilitating Academic Dishonesty: Intentionally or knowingly helping or attempting to help another to violate any provision of this Code. • Plagiarism: Intentionally or knowingly representing the words or ideas of another as one's own in any academic exercise.

교과정책(방침)

4C (Clear, Confident, Critical and Creative)